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# Benchmarks and Tests of a Multidimensional Cluster Dynamics Model of Helium Implantation in Tungsten

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**Abstract** — We present a hierarchical multiscale modeling study of implanted helium (He) segregation near grain boundaries (GBs) of tungsten. We extend our spatially dependent cluster dynamics model to two spatial dimensions in order to take into account the biased drift of mobile He clusters toward the GBs observed in atomic-scale simulations. We are able to reproduce the results from large-scale molecular dynamics simulations near and away from the GBs at low fluence with the extended cluster dynamics model. We suggest and verify that the sink (surface and GB) strengths are attenuated by the increasing concentration of He clusters at high fluence. This cluster dynamics model continues to set the stage for development of fully atomistically informed, coarse-grained models for computationally efficient predictions of He retention and surface morphological evolution, advancing progress toward the goal of efficient and optimal design of plasma-facing components.

**Keywords** — Cluster dynamics simulation, grain boundaries, plasma-exposed tungsten.

**Note** — Some figures may be in color only in the electronic version.

## I. INTRODUCTION

Understanding and predicting material damage in plasma-facing components (PFCs) is a challenge for next-generation tokamak fusion reactors like ITER and DEMO. Tungsten (W) is currently the material of choice for the divertor, the purpose of which is to extract heat produced by the fusion reaction. Tungsten has been chosen due to its high thermal conductivity, low sputtering yield, and high melting point.<sup>1</sup> Studies performed in either a linear plasma device or tokamak plasma show that helium (He)

irradiation causes the formation of microscopic features known as “fuzz” or “coral”<sup>2–8</sup> that can ultimately lead to erosion, degradation of material performance, and material failure. Even though this fuzz formation has been linked to the dynamics of He atoms and aggregates near the PFC surface, its exact mechanism remains unclear.

Different computational techniques can be used to study the effect of He plasma exposure on W crystals. Molecular dynamics (MD) has been favored in recent years as it allows the study of specific processes like bubble growth and bursting in W (Refs. 9 through 12), He migration kinetics and thermodynamics,<sup>13–16</sup> and the effects of surfaces<sup>17–21</sup> and grain boundaries<sup>21–24</sup> (GBs) on

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He transport. However, MD simulations are restricted to small spatial domains, using implantation fluxes orders of magnitude higher than experimental conditions ( $\sim 10^{25}$  He m $^{-2}$  s $^{-1}$  versus  $\sim 10^{20}$  He m $^{-2}$  s $^{-1}$ ), and only limited timescales of the order of 1  $\mu$ s can be reached.

Several approaches can be taken to overcome these limitations. Ito and coworkers<sup>25</sup> suggested to couple MD simulations to Monte Carlo random walk in order to accelerate the diffusion of He and reach longer timescales. Another successful approach is to build a continuum reaction-diffusion model in which parameters are informed from MD simulations like in Refs. 26 and 27 that allows one to extend the spatial domain, to use experimental flux values, and easily reach longer timescales.

In this paper, we continue our demonstration of a new continuum-scale drift-diffusion-reaction (DDR) cluster dynamics software, Xolotl,<sup>28</sup> for modeling the surface response of plasma-exposed W. Xolotl is being developed within a hierarchical multiscale modeling framework, in which we recently documented the ability to model surface segregation and He clustering behavior of 100 eV He implantation in single-crystal W using a one-dimensional (1-D) version of Xolotl as a function of surface orientation.<sup>18</sup> Here, we describe an extension to multiple spatial dimensions and include GBs as part of a more realistic representation of polycrystalline W. This extension to multiple spatial dimensions is a major stepping stone toward developing a fully resolved divertor simulator. We chose to specifically study the effect of the presence of GBs to showcase the multiple spatial dimensions aspect of Xolotl. Indeed, if GBs are not present in the domain, our multi-dimensional model would be equivalent to the 1-D one described in one of our previous papers<sup>18</sup> as well as in another paper in this special issue of *Fusion Science and Technology*.<sup>29</sup> Our benchmark strategy relies on incorporating mechanistic input from atomistic MD simulations and comparing to large-scale MD simulations of He agglomeration.

Large-scale MD simulations of He-irradiated W with the presence of GBs show that the behavior of He clusters is anisotropic.<sup>21</sup> Hu et al.<sup>22</sup> and Hammond et al.<sup>30</sup> have shown that the GBs act as sinks to mobile He clusters in the same manner as free surfaces, although He is effectively trapped on the GB rather than being released as it is for free surfaces. Previous studies characterized the behavior of small He clusters (composed of one to seven He atoms) near low-Miller-index free surfaces in W as well as near GBs using molecular statics (MS) and MD simulations.<sup>18,20,21,31,32</sup> This paper will focus on the implementation of multiple spatial dimensions and the incorporation of GBs in Xolotl, as well as benchmarks for comparison to a growing database of MD simulations. The use of this

kind of continuum methodology is necessary to predict material damage at fuzz-relevant spatial and temporal scales, unattainable with molecular-scale simulations.

First, in Sec. II.A we will describe the DDR rate theory-based cluster dynamics model Xolotl and our extensions to incorporate multiple spatial dimensions and GBs. Section II.B will briefly summarize the atomistic MD simulations used as benchmarks and to inform the continuum model. Section III will present and discuss the results of the benchmark studies of low-energy He implantation into W containing  $\Sigma 3$  symmetric tilt GBs. Finally, the most important conclusions of this work are summarized in Sec. IV.

## II. COMPUTATIONAL METHODS

### II.A. Cluster Dynamics Model

Xolotl predicts the evolution of the concentration fields of the various clusters in the material by solving the 1-D DDR equations with an incoming flux of He:

$$\begin{aligned} \partial_t C_i(z, t) = & -\nabla \cdot u_i(z) C_i(z, t) + D_i(T) \nabla^2 C_i(z, t) \\ & - Q_i(\bar{C}(z, t)) + \Gamma \cdot \rho_i(z, t) , \end{aligned} \quad (1)$$

where  $\bar{C}$  is the concentration vector, in which element  $C_i$  represents the volumetric concentration of cluster  $i$ . The first term on the right side represents the divergence of the flux due to drift, the second term comes from diffusion (Fick's first law), the third term is a generation term that includes all reactions involving cluster  $i$ , and the last term is another generation term arising from the implanted He flux  $\Gamma$  with  $\rho_i(z, t)$  the probability density function. The clusters under consideration in this work are of four types: W self-interstitials, W self-vacancies, He, and trapped He-vacancy (HeV) clusters; the size of each type can vary in a domain suggested by MD simulations.

Only small He clusters, interstitial clusters, and the single W vacancy move in this model. The diffusion term is isotropic, and the diffusion coefficient  $D_i$  depends strongly on temperature through an Arrhenius form,

$$D_i = D_{0,i} e^{-E_m/k_B T} , \quad (2)$$

where

$D_{0,i}$  = preexponential factor

$E_m$  = migration energy

$k_B$  = Boltzmann constant

$T$  = temperature.

The drift of mobile He clusters caused by surfaces and GBs has been studied by Hu et al.<sup>19–22</sup> and Hammond et al.<sup>30</sup> and is implemented here as a drift term toward the surface as well as toward GBs; the drift term for GBs is discussed later. The mobile vacancy and interstitial clusters are not subjected to drift in this work, although the framework exists to incorporate their biased drift.

The reaction term includes contributions from two types of reaction in which a cluster can be involved, namely, production and dissociation:



and



The theory of diffusion-limited reactions<sup>33</sup> gives the following rate coefficient for production reactions:

$$k^+ = k_{A,B}^+ = 4\pi(r_A + r_B)(D_A + D_B) \quad (5)$$

where  $r_A$  and  $r_B$  are the effective “capture” radii of  $A$  and  $B$ , respectively, and  $D_A$  and  $D_B$  are their respective diffusion coefficients in W.

The rate of the dissociation reaction (4) is given by

$$k^- = k_{G,H}^- = \frac{1}{\Omega} k_{G,H}^+ e^{\frac{-E_b}{k_B T}} \quad (6)$$

where  $\Omega$  is the atomic volume and  $E_b$  is the binding energy of the dissociating cluster ( $F$ , in this example).

The set of possible reactions is defined to be the same at every spatial grid point. The parameters needed in Eqs. (1), (5), and (6) are obtained from other simulations: MD simulations provide the range for the incoming He flux, the diffusion parameters, and the formation energies; the capture radii of a cluster is computed from geometric considerations.

The modifications to the finite difference implementation of Eq. (1) needed to use a grid with finer mesh size near the surface and coarser in the bulk are detailed in our previous work.<sup>18</sup> They are used here to get a better accuracy just below the surface. In order to study the effect of the GB with the Xolotl simulation code, DDR Eq. (1) needs additionally to be generalized to two dimensions. If we consider the biased drift only toward the surface, the drift term remains unchanged. Similarly, for a single crystalline material, the source term depends only on the depth and thus does not require updating. The reaction term is not modified either because it relies only on the local grid point and not on its neighbors. Finally, the diffusion term has to be generalized to two dimensions in the following way:

$$\begin{aligned} \frac{\partial C_i}{\partial t} = D_i & \left[ \frac{2}{a(a+b)} \left( C_i(x, z-a) \right. \right. \\ & + \frac{a}{b} C_i(x, z+b) - \left[ 1 + \frac{a}{b} \right] C_i(x, z) \Big) \\ & + \frac{C_i(x+h_x, z) + C_i(x-h_x, z) - 2C_i(x, z)}{h_x^2} \Big] \quad (7) \end{aligned}$$

where  $a$  and  $b$  are the grid sizes on the top and bottom sides, respectively, of the grid point located at  $(x, z)$ ;  $x$  is the coordinate along a direction parallel to the surface of the material that adds a new dimension to the problem. The grid in this direction has a constant spacing parameter  $h_x$ .

To complete our general model, we must include the surface growth phenomenon that implicitly occurs in atomistic MD simulations. This phenomenon takes the form of self-interstitial W atoms created during trap mutation reactions that are highly mobile and can reach the surface. There, they can either be ablated and leave the material or remain deposited and become adatoms. The accumulation of these adatoms on the surface generates new layers of material. To model this in Xolotl, a grid that extends beyond what is actually needed at the beginning of a simulation is used in order to leave room for growth, and the position of the surface becomes a variable that is explicitly tracked. The surface here is the limit between the vacuum region of the grid, where nothing happens (i.e., the DDR equations are not in effect), and the solid part in which He and vacancy clusters are tracked. During the simulation, Xolotl tracks the number of self-interstitial atoms that diffuses through the current boundary of the surface at a given grid point. When this number is consistent with an atomic density greater than the density of W, the surface position is moved upward by one grid point.

It has been shown by Hu et al.<sup>21</sup> and Hammond et al.<sup>30</sup> that GBs act as sinks for mobile He clusters in the same way that free surfaces do. The biased drift of the clusters has been parameterized with an elastic interaction potential

$$u_i(x) = \frac{3D_i A_i}{k_B T} \frac{1}{|x_{GB} - x|^4} \quad (8)$$

where

$D_i$  = Fick diffusion coefficient for cluster  $i$

$A_i$  = sink strength for cluster  $i$

$k_B$  = Boltzmann constant

$T$  = temperature

$x_{GB}$  =  $x$ -coordinate of the GB plane.

In the current implementation, the GB plane is coincident with the  $y$ - $z$  plane. This biased drift has been implemented in Xolotl as a drift term in the  $x$ -direction, which is parallel to the surface. This is only possible with a two-dimensional (2-D) version of the DDR equations.

Hammond et al.<sup>30</sup> also showed using MD simulations that once He clusters are on the GB, their diffusion is hindered. This required an extension to the drift and diffusion models in Xolotl from their original implementation. It was achieved by adding a Boolean grid to the code that has the same geometry as the spatial grid but only keeps the information of where mobile clusters are free to move (true) and where they no longer diffuse or drift (false, which occurs on the GB in our case).

## II.B. Atomistic Simulation

Numerous targeted MS simulations were performed to properly parameterize for implementation in Xolotl the biased drift of mobile He clusters toward a GB. The sink strengths in Eq. (8) are obtained from fits to simulation results as explained in the paper by Hu et al.<sup>22</sup> The simulations were conducted using the free MD software LAMMPS (Ref. 34), with the interactions between W atoms being described according to the Embedded Atom Method potential of Finnis and Sinclair<sup>35</sup> as modified initially by Ackland and Thetford<sup>36</sup> and more recently by Juslin and Wirth.<sup>37</sup> Tungsten-He pairwise interactions were described according to the pair potential of Juslin and Wirth,<sup>37</sup> and He-He interactions were described by the pair potential of Beck<sup>38</sup> as modified at short distances by Morishita and coworkers.<sup>39</sup> Small He clusters were inserted at random positions within 10 to 20 atomic layers of the plane of the GB. Conjugate-gradient energy minimization was then employed to calculate the energy of the relaxed atomic configuration with the embedded He cluster as a function of its distance away from the GB.

Moreover, a large-scale MD simulation containing two  $\Sigma 3\{111\}\{211\}$  symmetric tilt GBs, with the GB planes intersecting the surface and oriented normal to the surface plane at a distance of  $\sim 25$  nm from each other, has been performed and has provided a clear indication of He cluster segregation with the clusters accumulating near the surface and on the GBs (Ref. 21). For this simulation, the supercell size was  $\sim 50 \times 50 \times 21$  nm, where the top boundary was a free surface with a (111) crystallographic orientation. Periodic boundary conditions were applied in the  $x$ - and  $y$ -directions only, i.e., in the principal directions of the (horizontal) surface plane. The bottom few layers of W atoms were frozen at their crystallographic coordinates to reproduce a quasi-reflective image stress, which prevents He or W atoms from leaving through the bottom of

the simulation box. Helium atoms were inserted every 10 ps at random horizontal locations and at depths sampled from a depth distribution characteristic of that surface orientation at 933 K and 100-eV incident energy,<sup>31,32</sup> corresponding to a flux of  $4.0 \times 10^{25}$  He  $\text{m}^{-2} \text{s}^{-1}$ . This flux excludes He ions that reflect off the surface without embedding. This choice of the flux reflects the fact that MD simulations are limited to higher flux values, as well as small spatial and short temporal scales, in comparison to typical experimental studies. The simulation was also conducted using LAMMPS (Ref. 34), with the same interatomic potentials that were used in the MS simulations, for consistency.

## III. RESULTS AND DISCUSSION

Figure 1 shows a snapshot of the large-scale MD simulation described in Sec. II.B. The detailed reasons for the accumulation near the surface are described in our previous work,<sup>18,30–32</sup> but a succinct description of the processes happening near the free surface is useful for discussion purposes. Three main contributions to the surface segregation have been identified, all affecting mobile He clusters and depending on the surface orientation. The first one is the range of the incoming He atoms in the W material, the second one is the biased drift of mobile He clusters toward the surface, and the third one is the trap mutation near the surface due to He clusters that are too small to induce such reactions in the bulk.

In addition to the contribution of the drift that biases transport toward the GB, Hu et al.<sup>21,22</sup> also identified all of the reactions undergone by mobile He clusters near a  $\Sigma 3$  GB. Similar to the process near the surface, mobile He

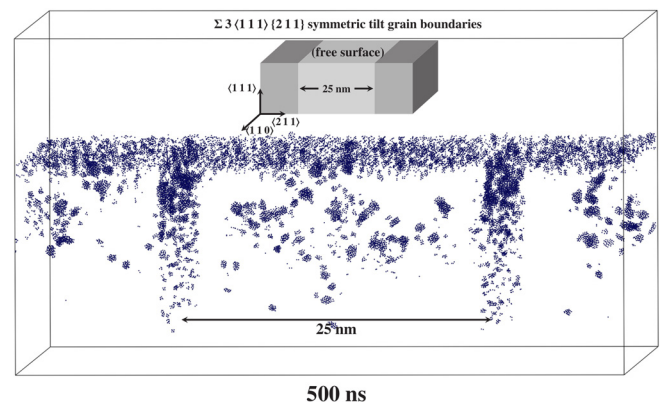


Fig. 1. Distribution of He atoms from large-scale MD simulation at 933 K after 500 ns, using polycrystalline W models with  $\Sigma 3\{111\}\{211\}$  symmetric tilt GBs at a distance of  $\sim 25$  nm from each other intersecting perpendicularly W (111) free surfaces with an incoming He flux of  $4.0 \times 10^{25}$  He  $\text{m}^{-2} \text{s}^{-1}$ . The inset gives a schematic representation of the MD supercell used in the simulation.



clusters near the GB can induce trap mutation at rates higher (and with smaller clusters) than in the bulk. These reactions are not listed in any detail here because they are not yet implemented in our cluster dynamics model. As described in Sec. II, the mobile He clusters are simply immobilized once they reach the GB, where they can then undergo trap mutation at the same rate as they do in the bulk of the material.

Large-scale MD simulations were used as a benchmark for the effects of GBs in the Xolotl simulation. However, because of the different types of simulation methods (atomistic/continuous time versus coarse grained/discrete time), a few distinctions have to be noted. First, in Xolotl, we limited the simulation to two dimensions as the third one is redundant by translational symmetry with the boundary conditions we are studying. The grid size used by Xolotl is 0.1 nm near the surface, while a variable grid size is used in the depth direction (with a finer mesh near the surface and coarser in the bulk). Second, the superposition of three drift fluxes is actually used in Xolotl: one toward the surface as discussed in Sec. II.A and two toward the two GBs using Eq. (8). These drift rates are determined from elastic inclusion theory as parameterized by MS simulations.<sup>21</sup>

Figure 2 shows the total concentration of He as a function of depth in a 4-nm-wide zone framing each GB at a fluence of  $2 \times 10^{19}$  He m<sup>-2</sup> in three cases: (1) absence of GBs in the computational domain, (2) drift toward the GBs but with He clusters that are still mobile when on the GBs, and (3) combination of drift with the immobilization of the clusters on the GBs. It can clearly be seen that the lack of mobility is important to the concentration profile around the GB, as the presence of drift alone is not enough to get a significantly different distribution from the case where no GBs are present in the simulation. Differences in the shapes of the profiles can be explained by considering the modified trap mutation process happening near the surface and the shape of the implanted He distribution. The latter peaks around 1 nm below the (111) surface of a W crystal, which is close to what is observed on the smooth part of the solid-circle data points. Mobile He clusters are attracted to sinks such as surfaces, which results in a lower concentration of He in the vicinity of the sink. They become trapped on their way to the surface through trap mutation processes, resulting in the peak in He concentration near 1 nm below the surface. This also explains why this peak in the He concentration profile of Fig. 2 is higher when He clusters are mobile compared to when they are immobile.

An important test of the parameterization of the model is to compare the results to MD simulation results. Figures 3 and 4 show the total He concentration around

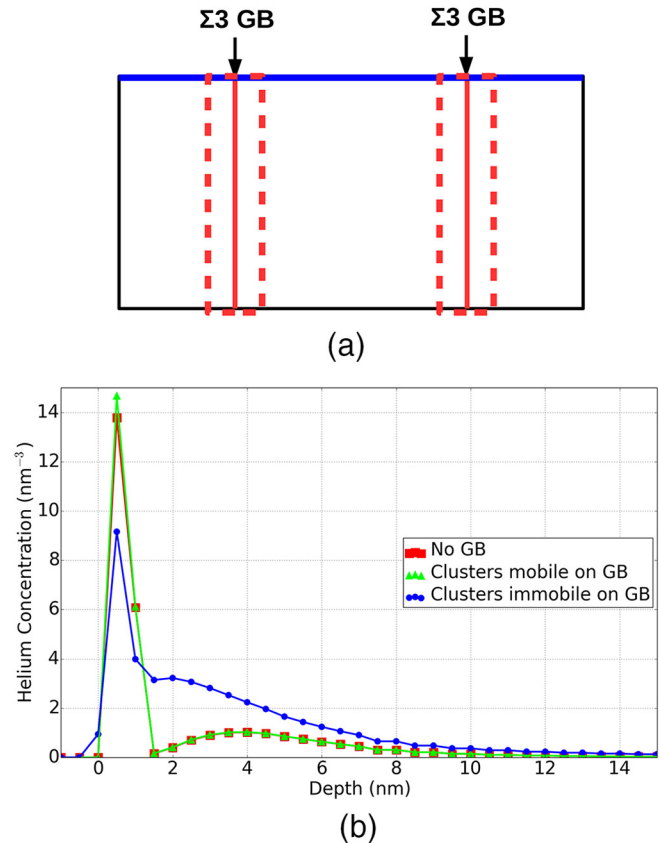


Fig. 2. (a) Schematic view of the 2-D cluster dynamics computational domain where the zones framed by the dotted lines correspond to where we integrate the He concentration,  $\pm 2$  nm from the GBs. (b) Total He concentration as a function of depth in these zones. Solid squares, triangles, and circles correspond to Xolotl simulation results with no GBs present, with biased drift toward the GBs but with the He clusters remaining mobile on the GBs, and with biased drift toward the GBs with the clusters becoming immobile once on the GBs, respectively.

the GBs and away from the GBs, respectively, as a function of depth for Xolotl and MD results at two fluences. It can be noticed that the Xolotl predictions for the total He concentration away from the GBs are similar to those of Fig. 2 in the case where no GBs are present in the domain. This is explained by the fact that the same processes are at play: trap mutation near the surface and drift diffusion toward the surface. Comparison of the Xolotl and MD data points demonstrates that the model implemented in Xolotl works well at low fluence, but when the fluence gets higher (above  $\sim 10^{19}$  He m<sup>-2</sup>), the discrepancies between the model predictions and the MD simulation results are important.

To understand these discrepancies, we have to return to how the parameters are obtained. The main difference in the distributions is the peak value of the He concentration located near the surface of the material. This

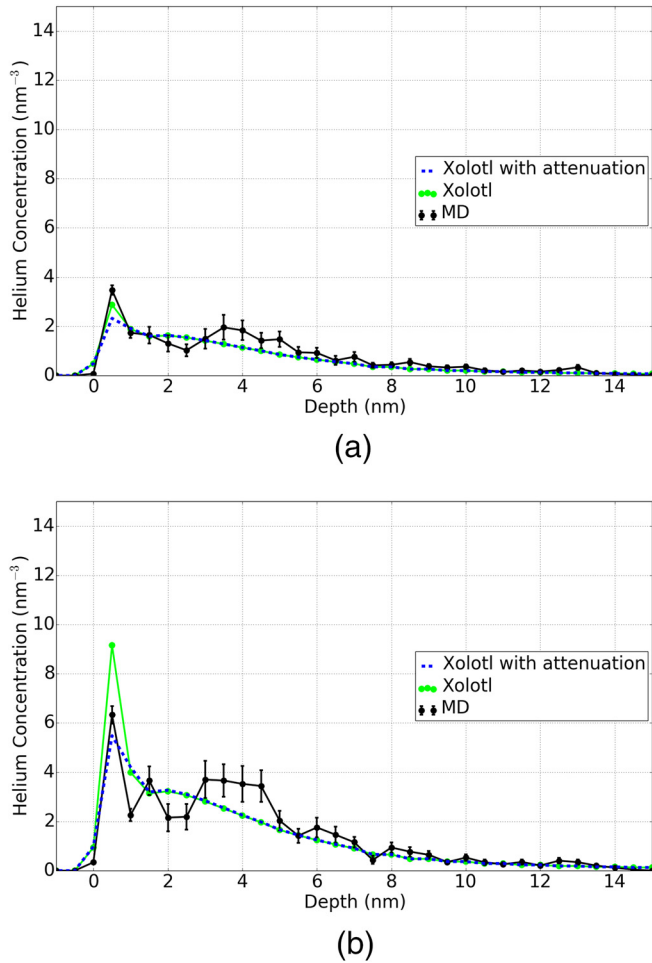


Fig. 3. Total He concentration as a function of depth near the GBs ( $\pm 2$  nm as on Fig. 2). The solid circles, dotted line, and solid circles with error bars represent results from a Xolotl simulation with GBs in the computational domain, results from the same simulation but with an attenuated trap mutation rate near the surface, and results from MD simulation with the error bars denoting the corresponding statistical errors, respectively. The fluence is (a)  $1 \times 10^{19}$  He  $\text{m}^{-2}$  and (b)  $2 \times 10^{19}$  He  $\text{m}^{-2}$ .

peak is due to the modified trap mutation that has been introduced near the surface. However, as described in our previous work,<sup>18</sup> the trap mutation of small He clusters is studied with MD simulations for a small volume of W material where only one He cluster is implanted 10 to 20 atomic layers below the surface in each simulation. This means that all the parameters that we use are valid only within the dilute limit of the He cluster concentration in the material, which is true only at lower fluence.

One hypothesis that could solve this problem without having to obtain new parameters for every He concentration is that the interaction strength of the sink (surface or GB) becomes gradually reduced when He atoms

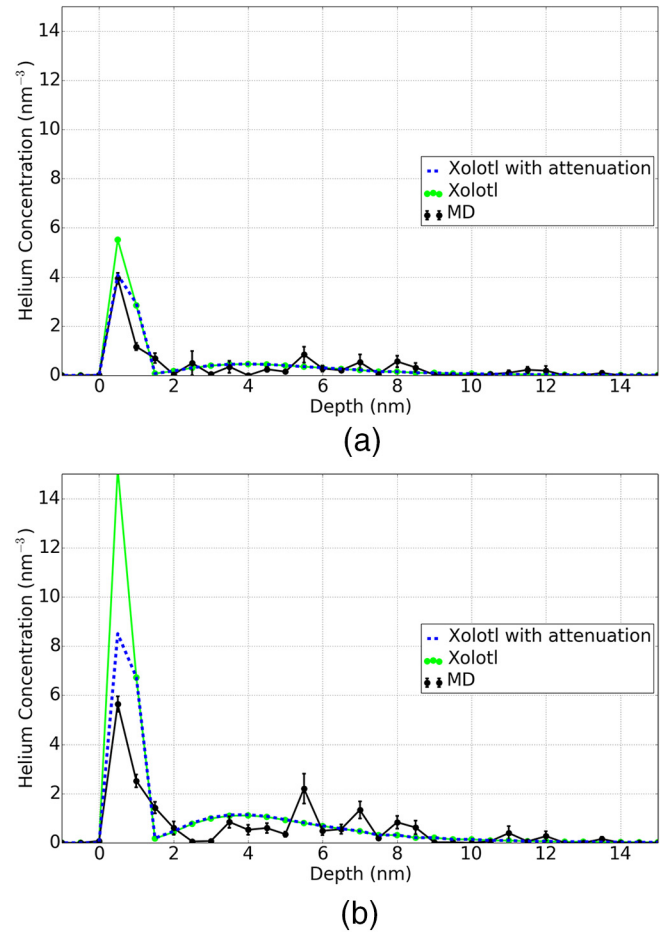


Fig. 4. Total He concentration as a function of depth away from the GBs. The solid circles, dotted line, and solid circles with error bars represent results from a Xolotl simulation with GBs in the computational domain, results from the same simulation but with an attenuated trap mutation rate near the surface, and results from MD simulation with the error bars denoting the corresponding statistical errors, respectively. The fluence is (a)  $10^{19}$  He  $\text{m}^{-2}$  and (b)  $2 \times 10^{19}$  He  $\text{m}^{-2}$ .

accumulate on it because it is no longer a free surface. This can be modeled by implementing an exponentially decreasing factor depending on the local He concentration in the modified trap mutation rate or the sink strength, for instance,

$$k'_{TM} = k_{TM} \cdot e^{-A \cdot C_{\text{He}}}, \quad (9)$$

where

$k'_{TM}, k_{TM}$  = updated and nominal trap mutation rates, respectively

$C_{\text{He}}$  = local He concentration

$A$  = adjustable parameter.

The dotted lines in Figs. 3 and 4 show the new results obtained with Xolotl in which the trap mutation rates near the surface are attenuated. The agreement is improved at higher fluence.

Looking at the concentration-versus-depth profiles from Xolotl and MD data for the He concentration near the GBs (Fig. 5), we can see on the Xolotl side a darker (higher concentration) zone of clusters containing 100 He atoms at a depth of 10 nm to clusters containing 40 He atoms at 20 nm. This zone actually continues outside of the limits of Fig. 5b and shows clusters containing 400 He atoms at a depth of 4 nm. If we now consider the MD results, the equivalent zone shows small cluster sizes at the same depth. The same is observed in the trap mutation peak near the surface, where Xolotl predictions show clusters bigger than those observed in the MD simulation. These are indications that Xolotl overestimates the growth

rate of smaller clusters, thus predicting larger sizes, as these zones are located more to the left (smaller clusters) in the MD results compared to the Xolotl predictions.

To determine whether this is due to the GB model or a problem related to the rate theory formulation itself, one can examine the same variables away from the GBs as shown in Fig. 6. The same problem appears in the peak located near the surface, meaning that this is not just related to our GB model. It could, however, imply that the theory of diffusion-limited reactions, where the production rates are given by Eq. (5), does not apply to these conditions. This can be additionally supported by the observation that at low total He concentrations, for instance, in the bulk of the material away from the GBs (Fig. 6), the discrepancy between MD and Xolotl results is not obvious. It should be noted, however, that the MD results are based on a single simulation—even though it is

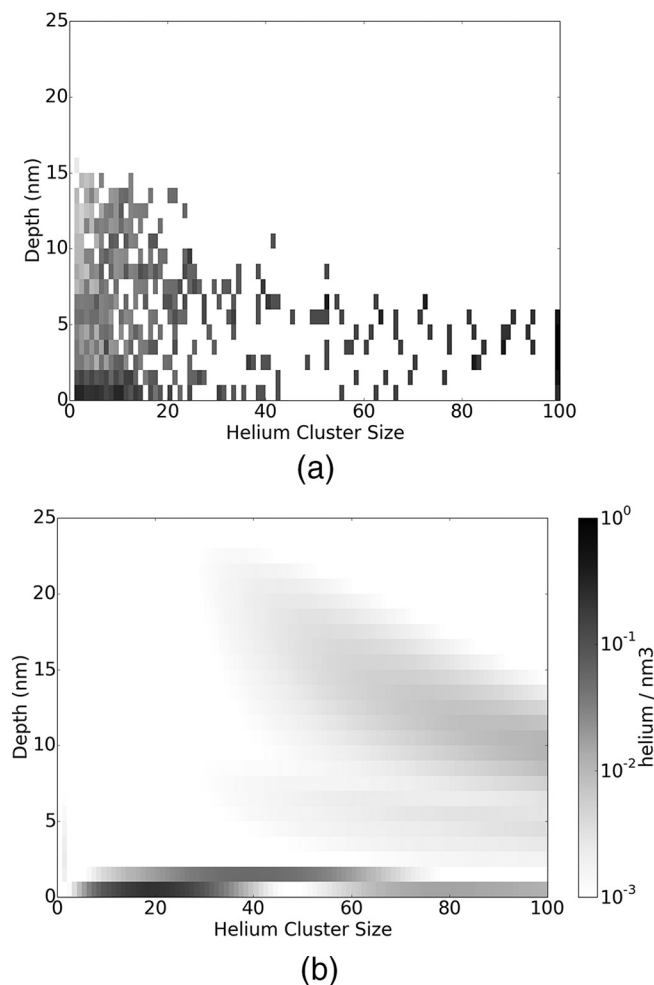


Fig. 5. Helium concentration as a function of depth and number of He atoms contained by the cluster near the GBs ( $\pm 2$  nm as in Fig. 2) at a fluence of  $4 \times 10^{19}$  He  $\text{m}^{-2}$ . (a) MD simulation results and (b) Xolotl modeling predictions.

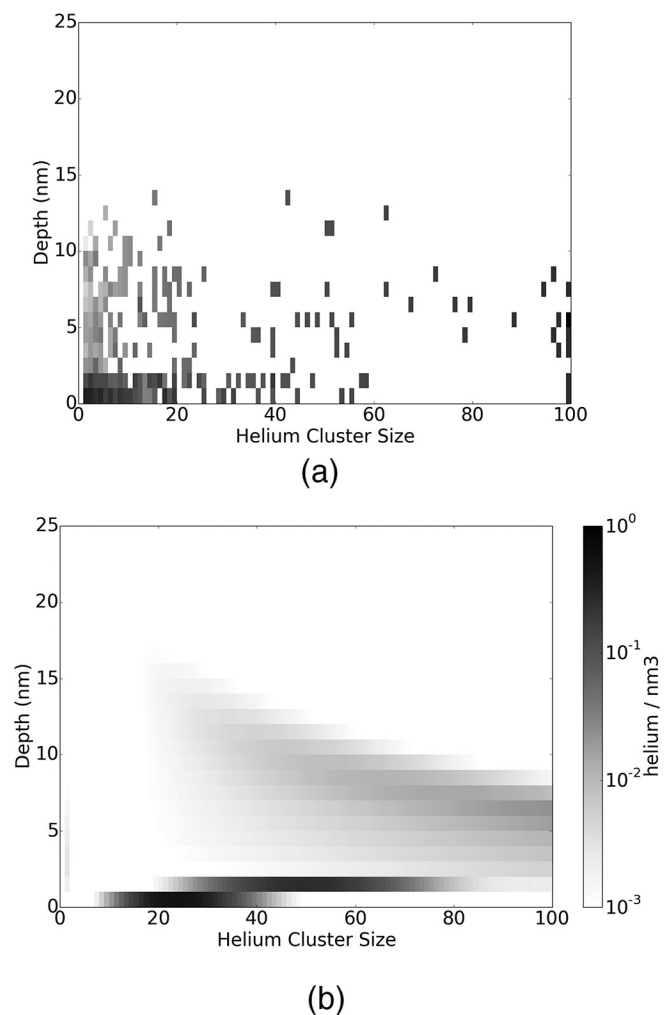


Fig. 6. Helium concentration as a function of depth and number of He atoms contained by the cluster away from the GBs at a fluence of  $4 \times 10^{19}$  He  $\text{m}^{-2}$ . (a) MD simulation results and (b) Xolotl modeling predictions.



a relatively large, and long, simulation—so there is plenty of room for statistical error in these observations.

More work is required to investigate this problem of overprediction of He cluster size, which is well beyond the scope of this paper, but revisiting the production rate definition will be a focus of our future work because of its important role in accurately capturing cluster dynamics at higher fluences within the Xolotl methodology.

#### IV. SUMMARY AND FUTURE WORK

To recapitulate, we extended our DDR rate theory-based cluster dynamics model, Xolotl, to two spatial dimensions and implemented the biased drift of mobile He clusters toward surfaces and GBs and compared this model to large-scale MD simulations. The parameterization employed in our work was obtained from systematic studies based on MS and MD simulations. We showed that our model is able to reproduce results from large-scale MD simulation with good agreement at low fluence, but the drift parameterization needs to be extended with an attenuation factor to match MD results at higher fluence.

The next step in this study is to add the modified trap mutation occurring because of mobile He clusters near GBs to our cluster dynamics model and explore how the results will be affected.

The fact that we are overpredicting the size of the HeV clusters in Xolotl has provided an insight into how the theory of diffusion-limited reactions could be preventing our model from accurately capturing cluster dynamics at high fluence. This issue may be a more general problem of the cluster dynamics method that will require substantial efforts for it to be addressed.

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